AI Calling Agent

Three ways the call is spoken

Phone today, localhost today, and the four-hour change proposed for Version7

Version4 is the Twilio phone path. Version6 is the localhost page in Cursor, and it stays frozen. Version7 is the copy where the next change would be made. This table includes only that change. It does not include a duplex WebSocket, streaming speech recognition, or a 350 millisecond full turn.

Architecture comparison

Architecture dimension	Phone path (Version4)	Localhost baseline (Version6)	Proposed in Research Papers (Version7)
Where it runs	Twilio app. The call opens /voice.	Cursor at http://127.0.0.1:8013/	Cursor at http://127.0.0.1:8014/
How audio moves	Twilio webhooks: /voice and /handle-speech.	HTTP: /local/listen, /local/turn, /local/speak, and /local/end.	The same HTTP calls. The first clause is spoken before the rest of the reply is synthesized.
How the customer is captured	Twilio Gather, Indian English, speech model experimental_utterances.	The browser records one WAV clip and uploads it.	The same WAV upload.
Speech to text	The text Twilio Gather returns.	Sarvam Saaras v4, one request for the whole clip.	The same Saaras request.
How a turn runs	One customer utterance, then one reply.	One customer utterance, then one full reply.	One customer utterance, then the reply is spoken clause by clause.
Interruption	Gather collects the next utterance.	The microphone stays open during playback and can stop the current clip.	The same interrupt, and clauses that have not played are dropped.

Architecture dimension	Phone path (Version4)	Localhost baseline (Version6)	Proposed in Research Papers (Version7)
Who writes the reply	Sarvam 105B, or a fixed line.	Sarvam 105B, or a fixed line. The full text is ready before speech starts.	The same full text. Speech then starts at the first clause.
Who speaks	Polly Kajal-Neural, the whole line.	Bulbul v3 Ishita, the whole clip. “sq. ft.” is spoken as “square feet.” Neerja speaks only if Bulbul returns nothing.	The same voice and wording, starting with the first clause.
How you hear it	Twilio plays the audio on the phone.	The browser plays one complete audio file.	The browser plays the first clause, then the following clauses.
Timing	A live phone turn was not measured for this table.	One measured localhost turn was about 3.8 seconds for the whole turn: 1.0 second recognition, 0.8 second model, 2.3 seconds for the full voice clip.	The wait before the first words can shrink toward the first clause. Recognition and model time stay.

What the Version7 change would achieve

Ishita can start the answer before the rest of the sentence is synthesized. If the tester speaks over her, the clauses still waiting do not continue. The Townpark stages, the project facts, the Opus 5.5 scorecard, and the Polly phone voice stay as they are.

The caller remains Sarvam 105B. Speech recognition remains one Saaras request for the finished clip. The phone customer still hears Polly Kajal-Neural.